

Current live architecture mapping for the Curriculum-Aware Academic Planning System, running entirely on serverless cloud infrastructure and client-side database connections.

CLIENT APPLICATION (FRONTEND)

React & TypeScript

v19.2 / v5.9

Client framework driving responsive component layouts, unified custom hooks, and safe static typing.

Vite

v7.3

Next-generation development server and build bundler optimizing asset sizes.

Tailwind CSS (v4) & PostCSS

v4.2

Utility-first modern styling setup utilizing native CSS variables and lightning-fast compilation.

Recharts

v3.7

SVG component charts rendering analytics dashboards, study progression, and score tracking.

Framer Motion

v12.34

Physics-based spring animation engine animating page views and interactive state indicators.

jsPDF

v4.2

Client-side document generator producing downloadable PDF files from user flashcard quizzes.

SERVERLESS BACKEND (CLOUD SERVICES)

Supabase JS Client SDK

v2.42

Direct client connection SDK for authenticating users, managing table queries, and invoking serverless edge processes.

Supabase Edge Functions

Deno Runtime

Serverless environment hosting core functions like `generate-plan` and `generate-quiz`.

Google Gemini 2.5 Flash API

REST API Integration

Direct HTTPS REST integration generating smart MCQs using user mistake history contexts.

Allocation & Risk Engines

Custom Deno Scripts

Deno-based scheduler resolving `PriorityScores` and `RiskFactors` based on course weights and study sessions.

DATABASE & IDENTITY MANAGEMENT

Supabase PostgreSQL DB

PostgreSQL v15+

Cloud database schema managing `User`, `AcademicProfile`, `UserCourse`, `UserTopic`, `StudyPlan`, `StudySession`, `ProgressLog`, `QuizResult`, and `MistakeLog`.

Database Triggers & Synchronization

PL/pgSQL Trigger

Database trigger (`user_sync_trigger.sql`) keeping Supabase auth users synchronized automatically with the public profiles.

Supabase Auth

GoTrue Identity

Handles user signups, sign-ins, session persistence, token issues, and OAuth routing.

Deployment

Vercel Config

Production hosting config for routing client calls and handling environmental scopes.